

Team 5 Presentation Bowowa

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$$F = -kx = m \frac{d^2 x}{dt^2} \quad (1)$$

$$x(t) = A \sin(\omega t) + B \cos(\omega t) \quad (2)$$

$$\omega \equiv \sqrt{\frac{k}{m}} \quad (3)$$

$$V(x) = \frac{1}{2} m \omega^2 x^2 \quad (4)$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + \frac{1}{2} m \omega^2 x^2 \psi = E \psi \quad (5)$$

1

2

(5)

$$\frac{1}{2m} [p^2 + (m\omega x)^2] \psi = E\psi \quad (6)$$

$$H = \frac{1}{2m} [p^2 + (m\omega x)^2] \quad (7)$$

$$a_{\pm} \equiv \frac{1}{\sqrt{2\hbar m\omega}} (\mp ip + m\omega x) \quad (8)$$

$$a_- a_+$$

$$\begin{aligned} a_- a_+ &= \frac{1}{2\hbar m\omega} (ip + m\omega x)(-ip + m\omega x) \\ &= \frac{1}{2\hbar m\omega} [p^2 + (m\omega x)^2 - im\omega(xp - px)] \end{aligned} \quad (9)$$

$$(xp - px), \quad x p \quad A B$$

$$[A, B] = AB - BA \quad (10)$$

$$(9) \quad a_- a_+ = \frac{1}{2\hbar m\omega} [p^2 + (m\omega x)^2] - \frac{i}{2\hbar} [x, p] \quad (11)$$

$$x p \quad f(x)$$

$$[x, p] f(x) = \left[x \frac{\hbar}{i} \frac{d}{dx} (f) - \frac{\hbar}{i} \frac{d}{dx} (x f) \right] = \frac{\hbar}{i} \left(x \frac{df}{dx} - x \frac{df}{dx} - f \right) = i \hbar f(x) \quad (12)$$

$$[x, p] = i \hbar \quad (13)$$

(11)

$$a_- a_+ = \frac{1}{\hbar \omega} H + \frac{1}{2} \quad (14)$$

a_+

$$a_+ a_- = \frac{1}{\hbar \omega} H - \frac{1}{2} \quad (15)$$

$$[a_-, a_+] = 1 \quad (16)$$

(14)(15)

$$H = \hbar\omega(a_-a_+ - \frac{1}{2}) \quad (17)$$

$$H = \hbar\omega(a_+a_- + \frac{1}{2}) \quad (18)$$

$a_{\pm},$

$$\hbar\omega(a_{\pm}a_{\mp} \pm \frac{1}{2})\psi = E\psi \quad (19)$$

$$\psi \quad E \quad H\psi = E\psi, \quad a_+\psi \quad (E + \hbar\omega) \quad H(a_+\psi) = (E + \hbar\omega)(a_+\psi)$$

$$\begin{aligned}
 H(a_+\psi) &= \hbar\omega(a_+a_- + \frac{1}{2})(a_+\psi) \\
 &= \hbar\omega(a_+a_-a_+ + \frac{1}{2}a_+)\psi \\
 &= \hbar\omega a_+(a_-a_+ + \frac{1}{2})\psi \\
 &= a_+ \left[\hbar\omega(a_+a_- + 1 + \frac{1}{2})\psi \right] \\
 &= a_+(H + \hbar\omega)\psi = a_+(E + \hbar\omega)\psi = (E + \hbar\omega)(a_+\psi)
 \end{aligned} \tag{20}$$

$$a_-\psi \quad (E - \hbar\omega)$$

$$a_{\pm}$$

$$a_- \psi_0 = 0 \quad (21)$$

$$\psi_0(x)$$

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2} \quad (22)$$

$$(19) \quad \hbar\omega(a_+a_- + \frac{1}{2})\psi_0 = E_0\psi_0,$$

$$E_0 = \frac{1}{2}\hbar\omega \quad (23)$$

$$\psi_0,$$

$$\psi_n(x) = A_n(a_+)^n \psi_0(x)$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad (24)$$

$$A_n$$

$$a_{\pm} \psi_{n\pm 1},$$

$$a_+ \psi_n = c_n \psi_{n+1} \quad , \quad a_- \psi_n = d_n \psi_{n-1} \quad (25)$$

$$f(x) \quad g(x)$$

$$\int_{-\infty}^{+\infty} f^*(a_{\pm} g) dx = \int_{-\infty}^{\infty} (a_{mp} f)^* g dx \quad (26)$$

$$a_{\mp} \quad a_{\pm}$$

$$\begin{aligned} \int_{-\infty}^{+\infty} f^*(a_{\pm} g) dx &= \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{+\infty} f^* \left(\mp \hbar \frac{d}{dx} + m\omega x \right) g dx \\ &= \frac{1}{\sqrt{2\hbar m\omega}} \int_{-\infty}^{+\infty} \left[\left(\pm \hbar \frac{d}{dx} + m\omega x \right) f \right]^* g dx \\ &= \int_{-\infty}^{+\infty} (a_{\mp} f)^* g dx \end{aligned} \quad (27)$$

$$\int_{-\infty}^{+\infty} (a_{\pm}\psi_n)^*(a_{\pm}\psi_n)dx = \int_{-\infty}^{+\infty} (a_{\mp}a_{\pm}\psi_n)^*\psi_ndx \quad (28)$$

(19) (24)

$$a_+a_-\psi_n = n\psi_n \quad , \quad a_-a_+\psi_n = (n+1)\psi_n \quad (29)$$

$$\int_{-\infty}^{+\infty} (a_+\psi_n)^*(a_+\psi_n)dx = |c_n|^2 \int_{-\infty}^{+\infty} |\psi_{n+1}|^2 dx = (n+1) \int_{-\infty}^{+\infty} |\psi_n|^2 dx$$

$$\int_{-\infty}^{+\infty} (a_-\psi_n)^*(a_-\psi_n)dx = |d_n|^2 \int_{-\infty}^{+\infty} |\psi_{n-1}|^2 dx = n \int_{-\infty}^{+\infty} |\psi_n|^2 dx \quad (30)$$

$c_n d_n,$

$$a_+ \psi_n = \sqrt{n+1} \psi_{n+1} \quad a_- \psi_n = \sqrt{n} \psi_{n-1} \quad (31)$$

$$\psi_1 = a_+ \psi_0$$

$$\psi_2 = \frac{1}{\sqrt{2}} a_+ \psi_1 = \frac{1}{\sqrt{2}} (a_+)^2 \psi_0$$

$$\psi_3 = \frac{1}{\sqrt{3}} a_+ \psi_2 = \frac{1}{\sqrt{3 \times 2}} (a_+)^3 \psi_0$$

$$\psi_4 = \frac{1}{\sqrt{4}} a_+ \psi_3 = \frac{1}{\sqrt{4 \times 3 \times 2}} (a_+)^4 \psi_0$$

$$\psi_n = \frac{1}{\sqrt{n!}} (a_+)^n \psi_0 \quad (32)$$

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2\psi = E\psi$$

$$\xi \equiv \sqrt{\frac{m\omega}{\hbar}}x \quad (33)$$

$$\frac{d^2\psi}{d\xi^2} = (\xi^2 - K)\psi \quad (34)$$

$$K \equiv \frac{2E}{\hbar\omega} \quad (35)$$

ξ K

$$\frac{d^2\psi}{d\xi^2} \approx \xi^2\psi \quad (36)$$

$$\psi(\xi) \approx Ae^{-\frac{\xi^2}{2}} + Be^{\frac{\xi^2}{2}} \quad (37)$$

B

$$\psi(\xi) = h(\xi)e^{-\frac{\xi^2}{2}} \quad (38)$$

$$\begin{aligned}\frac{d\psi}{d\xi} &= \left(\frac{dh}{d\xi} - \xi h\right)e^{-\frac{\xi^2}{2}} \\ \frac{d^2\psi}{d\xi^2} &= \left(\frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (\xi^2 - 1)h\right)e^{-\frac{\xi^2}{2}} \\ \frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (K - 1)h &= 0\end{aligned}\tag{39}$$

ξ

$$h(\xi) = a_0 + a_1\xi + a_2\xi^2 + \cdots = \sum_{j=0}^{\infty} a_j \xi^j \quad (40)$$

$$\frac{dh}{d\xi} = a_1 + 2a_2\xi + 3a_3\xi^2 + \cdots = \sum_{j=0}^{\infty} j a_j \xi^{j-1}$$

$$\frac{d^2 h}{d\xi^2} = 2a_2 + 2 \times 3a_3\xi + 3 \times 4a_4\xi^2 + \cdots = \sum_{j=0}^{\infty} (j+1)(j+2)a_{j+2}\xi^j$$

(39)

$$\sum_{j=0}^{\infty} [(j+1)(j+2)a_{j+2} - 2ja_j + (K-1)a_j] \xi^j = 0 \quad (41)$$

ξ

$$(j+1)(j+2)a_{j+2} - 2ja_j + (K-1)a_j = 0$$

$$a_{j+2} = \frac{(2j+1-K)}{(j+1)(j+2)}a_j \quad (42)$$

 a_0

$$a_2 = \frac{(1-K)}{2}a_0, a_4 = \frac{5-K}{12}a_2 = \frac{(5-K)(1-K)}{24}a_0, \dots$$

 a_1

$$a_3 = \frac{(3-K)}{6}a_1, a_5 = \frac{7-K}{20}a_3 = \frac{(7-K)(3-K)}{120}a_1, \dots$$

$$h(\xi) = h_{\text{even}}(\xi) + h_{\text{odd}}(\xi) \quad (43)$$

$$h_{\text{even}}(\xi) \equiv a_0 + a_2\xi^2 + a_4\xi^4 + \dots$$

$$h_{\text{odd}}(\xi) \equiv a_1\xi + a_3\xi^3 + a_5\xi^5 + \dots$$

j

$$a_{j+2} \approx \frac{2}{j} a_j \quad (44)$$

$$a_j \approx \frac{C}{(j/2)!} \quad (45)$$

$$h(\xi) \approx C \sum \frac{1}{j!} \xi^j \approx C \sum \frac{1}{j!} \xi^{2j} \approx e^{\xi^2} \quad (46)$$

$$(38) \quad \psi = e^{\frac{\xi^2}{2}}, \quad j = n,$$

$$K = 2n + 1 \quad (47)$$

$$(35) \quad a_1 = 0, \quad a_0 = 0$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega \quad n = 1, 2, \dots$$

K ,

$$a_{j+2} = \frac{-2(n-j)}{(j+1)(j+2)} a_j \quad (48)$$

n=0

$$h_0(\xi) = a_0$$

$$\psi_0(\xi) = a_0 e^{-\frac{\xi^2}{2}}$$

n=1

$$h_1(\xi) = a_1 \xi$$

$$\psi_1(\xi) = a_1 \xi e^{-\frac{\xi^2}{2}}$$

n=2

$$h_2(\xi) = a_0(1 - 2\xi^2)$$

$$\psi_2(\xi) = a_0(1 - 2\xi^2) e^{-\frac{\xi^2}{2}}$$

$$H_n(\xi),$$

$$H_0 = 1$$

$$H_1 = 2\xi$$

$$H_2 = 4\xi^2 - 2$$

$$H_3 = 8\xi^3 - 12\xi$$

$$H_4 = 16\xi^4 - 48\xi^2 + 12$$

$$H_5 = 32\xi^5 - 160\xi^3 + 120\xi$$

(49)

$$H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi} \right)^n e^{-\xi^2} \quad (50)$$

$$H_{n+1}(\xi) = 2\xi H_n(\xi) - 2n H_{n-1}(\xi) \quad (51)$$

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\frac{\xi^2}{2}} \quad (52)$$